The Chain Rule

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Basic chain rule applications**

Differentiate each function using the chain rule:

a $f(x) = (4x - 1)^5$

$= 5(4x - 1)^4(4) = 20(4x - 1)^4$

b $f(x) = \sin(3x^2)$ $= 6x\cos(3x^2)$

c $f(x) = e^{5x}$ $= 5e^{5x}$

d $f(x) = \ln(2x + 7)$ $= \frac{2}{2x + 7}$

(a) $= 5(4x - 1)^4(4) = 20(4x - 1)^4$; (b) $= 6x\cos(3x^2)$; (c) $= 5e^{5x}$; (d) $= \frac{2}{2x + 7}$

2 **[Foundational]** • **Target: Basic chain rule applications**

Differentiate $f(x) = \sqrt{x^2 + 9}$.

$f'(x) = \frac{2x}{2\sqrt{x^2 + 9}} = \frac{x}{\sqrt{x^2 + 9}}$.

3 **[Foundational]** • **Target: The chain rule with trig and exponential functions**

Differentiate $f(x) = \cos^3(x)$ (i.e. $[\cos x]^3$).

$f'(x) = 3\cos^2 x \cdot (-\sin x) = -3\cos^2 x \sin x$.

4 **[Foundational]** • **Target: The chain rule with trig and exponential functions**

Differentiate $f(x) = e^{\sin x}$.

$f'(x) = \cos x \cdot e^{\sin x}$.

5 **[Foundational]** • **Target: The chain rule with trig and exponential functions**

Differentiate $f(x) = \tan(e^x)$.

$f'(x) = \sec^2(e^x) \cdot e^x$.

6 **[Foundational]** • **Target: Nested (multiple) chain rule**

Differentiate $f(x) = \sin\left(\sqrt{x^2 + 1}\right)$, applying the chain rule twice.

$f'(x) = \cos\left(\sqrt{x^2 + 1}\right) \cdot \frac{x}{\sqrt{x^2 + 1}}$.

7 **[Foundational]** • **Target: Nested (multiple) chain rule**

Differentiate $f(x) = (\ln(3x))^4$.

$f'(x) = 4(\ln(3x))^3 \cdot \frac{3}{3x} = \frac{4(\ln(3x))^3}{x}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Chain rule combined with product/quotient rules**

Differentiate $f(x) = x^2 e^{3x}$.

$f'(x) = 2xe^{3x} + x^2(3e^{3x}) = e^{3x}(2x + 3x^2)$.

9 **[Intermediate]** • **Target: Chain rule combined with product/quotient rules**

Differentiate $f(x) = \frac{\sin(2x)}{x}$.

$f'(x) = \frac{2\cos(2x) \cdot x - \sin(2x)}{x^2}$.

10 **[Intermediate]** • **Target: Chain rule with tables of values**

Given $f(2) = 5$, $f'(2) = 3$, $g(2) = 7$, $g'(2) = -1$, find $\frac{d}{dx}[f(g(x))]$ at $x = 2$ if $g(2) = 7$ and $f'(7) = 4$ (note the composition uses f' evaluated at $g(2) = 7$, not at 2).

$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$; at $x = 2$: $f'(7) \cdot g'(2) = 4 \cdot (-1) = -4$.

11 **[Intermediate]** • **Target: Chain rule with inverse trig and logarithms**

Differentiate $f(x) = \sin^{-1}(2x)$.

$f'(x) = \frac{2}{\sqrt{1 - (2x)^2}} = \frac{2}{\sqrt{1 - 4x^2}}$.

12 **[Intermediate]** • **Target: Chain rule with inverse trig and logarithms**

Differentiate $f(x) = \ln(\sin x)$.

$f'(x) = \frac{\cos x}{\sin x} = \cot x$.

13 **[Intermediate]** • **Target: Chain rule with inverse trig and logarithms**

Differentiate $f(x) = \tan^{-1}(x^2)$.

$f'(x) = \frac{2x}{1 + (x^2)^2} = \frac{2x}{1 + x^4}$.

14 **[Intermediate]** • **Target: Finding tangent lines using the chain rule**

Find the equation of the tangent line to $f(x) = (2x - 1)^3$ at $x = 1$.

$f(1) = 1^3 = 1$. $f'(x) = 3(2x - 1)^2(2) = 6(2x - 1)^2$; $f'(1) = 6(1)^2 = 6$. Tangent: $y - 1 = 6(x - 1)$, i.e. $y = 6x - 5$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Visualizing a composite function and its derivative • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

The graph shows $f(x) = (x-1)^2$ (an inner-outer composite of $u = x - 1$ and u^2), with its tangent line at $x = 2$.

a Use the chain rule to find $f'(x)$. $f'(x) = 2(x-1)(1) = 2(x-1)$.

b Confirm $f'(2)$ matches the slope of the tangent line shown. $f'(2) = 2(1) = 2$, matching the tangent line $y = 2x - 2$, which has slope 2.

(a) $f'(x) = 2(x-1)(1) = 2(x-1)$.; (b) $f'(2) = 2(1) = 2$, matching the tangent line $y = 2x - 2$, which has slope 2.

16 [Challenging] • Target: The chain rule in applied rate problems • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

The volume of a sphere is $V = \frac{4}{3}\pi r^3$, and the radius is growing according to $r(t) = 2t + 1$ (cm, seconds). Use the chain rule to find $\frac{dV}{dt}$ at $t = 2$.

$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt} = 4\pi r^2 \cdot 2$. At $t = 2$: $r = 5$, so $\frac{dV}{dt} = 4\pi(25)(2) = 200\pi \approx 628.3$ cm^3/s.

17 [Challenging] • Target: Chain rule with a table of composed values • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Given $h(x) = f(g(x))$ where $g(1) = 3$, $g'(1) = 2$, $f(3) = 7$, $f'(3) = -4$, find $h'(1)$.

$h'(x) = f'(g(x)) \cdot g'(x)$: $h'(1) = f'(g(1)) \cdot g'(1) = f'(3) \cdot 2 = (-4)(2) = -8$.

18 [Challenging] • Target: Chain rule with a table of composed values • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Differentiate $f(x) = \sqrt[3]{x^2 + 1}$ using the chain rule.

$f(x) = (x^2+1)^{1/3}$: $f'(x) = \frac{1}{3}(x^2+1)^{-2/3}(2x) = \frac{2x}{3(x^2+1)^{2/3}}$.

19 [Challenging] • Target: Chain rule with tables of values • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Given $g(6) = -5$, $g'(6) = -3$, and $f'(-5) = 6$, find $h'(6)$ if $h(x) = f(g(x))$.

Final Answer: -18. $h'(x) = f'(g(x)) \cdot g'(x)$, so $h'(6) = f'(g(6)) \cdot g'(6) = f'(-5) \cdot (-3) = (6)(-3) = -18$.

20 [Challenging] • Target: Chain rule with inverse trig and logarithms • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Given $g(2) = -5$, $g'(2) = 1$, and $f'(-5) = -5$, find $h'(2)$ if $h(x) = f(g(x))$.

Final Answer: -5. $h'(x) = f'(g(x)) \cdot g'(x)$, so $h'(2) = f'(g(2)) \cdot g'(2) = f'(-5) \cdot (1) = (-5)(1) = -5$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic chain rule applications

Use the chain rule to differentiate $f(x) = (x - 1)^2$, then find $f'(1)$.

a. 0 ✓

b. -2

c. 3

d. 5

$f'(x) = 2(x - 1)^1(1)$, so $f'(1) = 2(0)^1(1) = 0$.

22 [Foundational] • Target: The chain rule with trig and exponential functions

Use the chain rule to differentiate $f(x) = (4x - 3)^2$, then find $f'(1)$.

a. 4

b. 6

c. 2

d. 8 ✓

$f'(x) = 2(4x - 3)^1(4)$, so $f'(1) = 2(1)^1(4) = 8$.

23 [Foundational] • Target: Nested (multiple) chain rule

Use the chain rule to differentiate $f(x) = (4x - 1)^2$, then find $f'(1)$.

a. 24 ✓

b. 72

c. 12

d. 6

$f'(x) = 2(4x - 1)^1(4)$, so $f'(1) = 2(3)^1(4) = 24$.

24 [Foundational] • Target: Chain rule combined with product/quotient rules

Use the chain rule to differentiate $f(x) = (3x)^4$, then find $f'(1)$.

a. 81

b. 324 ✓

c. 108

d. 972

$f'(x) = 4(3x)^3(3)$, so $f'(1) = 4(3)^3(3) = 324$.

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25 [Foundational] • Target: Chain rule with tables of values

Use the chain rule to differentiate $f(x) = (2x + 3)^4$, then find $f'(0)$.

a. 108

b. 648

c. 216 ✓

d. 54

$f'(x) = 4(2x + 3)^3(2)$, so $f'(0) = 4(3)^3(2) = 216$.

26 [Foundational] • Target: Chain rule with inverse trig and logarithms

Use the chain rule to differentiate $f(x) = (x - 1)^3$, then find $f'(0)$.

a. 3 ✓

b. -3

c. 0

d. 1

$f'(x) = 3(x - 1)^2(1)$, so $f'(0) = 3(-1)^2(1) = 3$.

27 [Foundational] • Target: Finding tangent lines using the chain rule

Use the chain rule to differentiate $f(x) = (2x + 4)^2$, then find $f'(-2)$.

a. -2

b. 0 ✓

c. 2

d. 5

$f'(x) = 2(2x + 4)^1(2)$, so $f'(-2) = 2(0)^1(2) = 0$.

28 [Foundational] • Target: Visualizing a composite function and its derivative

Given $g(5) = 4$, $g'(5) = -2$, and $f'(4) = 5$, find $h'(5)$ if $h(x) = f(g(x))$.

a. 5

b. -2

c. 3

d. -10 ✓

$h'(x) = f'(g(x)) \cdot g'(x)$, so $h'(5) = f'(g(5)) \cdot g'(5) = f'(4) \cdot (-2) = (5)(-2) = -10$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: The chain rule in applied rate problems

Given $g(6) = -5$, $g'(6) = -6$, and $f'(-5) = -4$, find $h'(6)$ if $h(x) = f(g(x))$.

a. -6

b. 24 ✓

c. -4

d. -10

$h'(x) = f'(g(x)) \cdot g'(x)$, so $h'(6) = f'(g(6)) \cdot g'(6) = f'(-5) \cdot (-6) = (-4)(-6) = 24$.

30 [Intermediate] • Target: Chain rule with a table of composed values

Given $g(4) = -3$, $g'(4) = 6$, and $f'(-3) = 3$, find $h'(4)$ if $h(x) = f(g(x))$.

a. 9

b. 18 ✓

c. 6

d. 3

$h'(x) = f'(g(x)) \cdot g'(x)$, so $h'(4) = f'(g(4)) \cdot g'(4) = f'(-3) \cdot (6) = (3)(6) = 18$.

31 [Intermediate] • Target: Basic Chain Rule & Composite Functions

Given $g(7) = -4$, $g'(7) = 3$, and $f'(-4) = 5$, find $h'(7)$ if $h(x) = f(g(x))$.

a. 8

b. 5

c. 3

d. 15 ✓

$h'(x) = f'(g(x)) \cdot g'(x)$, so $h'(7) = f'(g(7)) \cdot g'(7) = f'(-4) \cdot (3) = (5)(3) = 15$.

32 [Intermediate] • Target: Exponential & Logarithmic Chain Rule

Given $g(3) = -5$, $g'(3) = 6$, and $f'(-5) = 2$, find $h'(3)$ if $h(x) = f(g(x))$.

a. 2

b. 6

c. 12 ✓

d. 8

$h'(x) = f'(g(x)) \cdot g'(x)$, so $h'(3) = f'(g(3)) \cdot g'(3) = f'(-5) \cdot (6) = (2)(6) = 12$.

33 [Intermediate] • Target: Repeated Chain Rule Applications

Given $g(3) = -6$, $g'(3) = -2$, and $f'(-6) = -3$, find $h'(3)$ if $h(x) = f(g(x))$.

a. -5

b. -3

c. 6 ✓

d. -2

$h'(x) = f'(g(x)) \cdot g'(x)$, so $h'(3) = f'(g(3)) \cdot g'(3) = f'(-6) \cdot (-2) = (-3)(-2) = 6$.

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34 [Intermediate] • Target: Table-Based Chain Rule Problems

Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 16}$, then find $f'(-3)$.

a. $-\frac{5}{3}$

b. $-\frac{3}{4}$

c. $-\frac{3}{5}$ ✓

d. $\frac{3}{5}$

$f'(x) = \frac{x}{\sqrt{x^2+16}}$, so $f'(-3) = \frac{-3}{\sqrt{(-3)^2+16}} = \frac{-3}{5} = -\frac{3}{5}$.

35 [Intermediate] • Target: Basic chain rule applications

Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 144}$, then find $f'(-5)$.

a. $-\frac{5}{12}$

b. $-\frac{13}{5}$

c. $-\frac{5}{13}$ ✓

d. $\frac{5}{13}$

$f'(x) = \frac{x}{\sqrt{x^2+144}}$, so $f'(-5) = \frac{-5}{\sqrt{(-5)^2+144}} = \frac{-5}{13} = -\frac{5}{13}$.

36 [Intermediate] • Target: The chain rule with trig and exponential functions

Given $h(2) = -3$, $h'(2) = -3$, $g(-3) = -1$, $g'(-3) = -4$, and $f'(-1) = -3$, find $k'(2)$ if $k(x) = f(g(h(x)))$.

a. -36 ✓

b. -10

c. 12

d. 9

$k'(x) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$, so

$k'(2) = f'(g(-3)) \cdot g'(-3) \cdot h'(2) = f'(-1) \cdot (-4) \cdot (-3) = (-3)(-4)(-3) = -36$.

37 [Intermediate] • Target: Nested (multiple) chain rule

Given $h(5) = 0$, $h'(5) = -5$, $g(0) = -5$, $g'(0) = 3$, and $f'(-5) = -4$, find $k'(5)$ if $k(x) = f(g(h(x)))$.

a. 60 ✓

b. -12

c. -6

d. 20

$k'(x) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$, so

$k'(5) = f'(g(0)) \cdot g'(0) \cdot h'(5) = f'(-5) \cdot (3) \cdot (-5) = (-4)(3)(-5) = 60$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Chain rule combined with product/quotient rules • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Given $h(2)=0$, $h'(2)=2$, $g(0)=1$, $g'(0)=-4$, and $f'(1)=-5$, find $k'(2)$ if $k(x)=f(g(h(x)))$.

a. 40 ✓

b. -10

c. 20

d. -7

$k'(x)=f'(g(h(x)))\cdot g'(h(x))\cdot h'(x)$, so

$k'(2)=f'(g(0))\cdot g'(0)\cdot h'(2)=f'(1)\cdot(-4)\cdot(2)=(-5)(-4)(2)=40$.

39 [Challenging] • Target: Chain rule with tables of values • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Which is the correct derivative of $F(x)=\sin(-2x-1)$?

a. $-2\cos(-2x-1)$ ✓

b. $2\sin(-2x-1)$

c. $-2\cos(x)$

d. $\cos(-2x-1)$

By the chain rule, $F'(x)=\cos(u)\cdot u'$ where $u=-2x-1$ and $u'=-2$: $F'(x)=-2\cos(-2x-1)$.

40 [Challenging] • Target: Chain rule with inverse trig and logarithms • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Which is the correct derivative of $F(x)=\sin(-6x+4)$?

a. $-6\cos(x)$

b. $-6\cos(-6x+4)$ ✓

c. $\cos(-6x+4)$

d. $6\sin(-6x+4)$

By the chain rule, $F'(x)=\cos(u)\cdot u'$ where $u=-6x+4$ and $u'=-6$: $F'(x)=-6\cos(-6x+4)$.

41 [Challenging] • Target: Basic Chain Rule & Composite Functions • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

If $h(x)=f(g(x))$ with $g(3)=5$, $g'(3)=-2$, and $f'(5)=7$, what is $h'(3)$?

a. -14 ✓

b. 10

c. -10

d. 35

By the chain rule: $h'(3)=f'(g(3))\cdot g'(3)=f'(5)\cdot(-2)=7\cdot(-2)=-14$.

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42 [Challenging] • Target: Exponential & Logarithmic Chain Rule • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Find the derivative of $f(x) = e^{5x^2 - 2x}$.

a. $(10x - 2)e^{5x^2 - 2x}$ ✓

b. $e^{10x - 2}$

c. $10xe^{5x^2 - 2x}$

d. $(5x^2 - 2x)e^{5x^2 - 2x}$

$\frac{d}{dx}[e^u] = u'e^u$. Here $u' = 10x - 2$, so $f'(x) = (10x - 2)e^{5x^2 - 2x}$.

43 [Challenging] • Target: Exponential & Logarithmic Chain Rule • Cross-Lesson: 8.01 Area Between Curves

Find $\frac{d}{dx}[\ln(\cos x)]$.

a. $-\tan x$ ✓

b. $\tan x$

c. $\cot x$

d. $-\cot x$

$\frac{1}{\cos x} \cdot (-\sin x) = -\frac{\sin x}{\cos x} = -\tan x$.

44 [Challenging] • Target: Repeated Chain Rule Applications • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Differentiate $f(x) = \sqrt{1 + \tan(3x)}$.

a. $\frac{3\sec^2(3x)}{2\sqrt{1 + \tan(3x)}}$ ✓

b. $\frac{\sec^2(3x)}{2\sqrt{1 + \tan(3x)}}$

c. $\frac{3\sec^2(3x)}{\sqrt{1 + \tan(3x)}}$

d. $\frac{1}{2\sqrt{1 + \tan(3x)}}$

$\frac{d}{dx}[\sqrt{u}] = \frac{u'}{2\sqrt{u}}$. Here $u' = 3\sec^2(3x)$.

45 [Challenging] • Target: Table-Based Chain Rule Problems • Cross-Lesson: 1.01 Limits Graphically and Numerically

Let $h(x) = [f(x)]^3$. If $f(1) = 2$ and $f'(1) = -4$, what is $h'(1)$?

a. -48 ✓

b. -24

c. 12

d. -16

$h'(x) = 3[f(x)]^2 f'(x) \Rightarrow h'(1) = 3(2)^2(-4) = 3(4)(-4) = -48$.

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